ISM – Mid Sem – EC2
Date: 19 Sep 2026

Question

You can write your answers in the provided space or write your answer on a piece of paper and scan and upload the handwritten answers using the QR Code available in the Scan and Upload Section of this exam.

Kindly ensure all answer sheets to be uploaded against the corresponding questions only. All sheets should not be uploaded against one question.

a). A hospital records ER waiting times (minutes) for 15 patients:
5, 8, 9, 12, 14, 15, 16, 18, 20, 22, 25, 28, 30, 45, 95
(i) Compute the five-number summary and IQR.
(ii) Identify any outliers using the 1.5×IQR rule.
(iii) Discuss whether the mean or median is a better measure of central tendency here, and why.

[3 Marks]

b). Among smartphone users, 65% use App A, 40% use App B, and 25% use neither app.
(i) Find P(use at least one app).
(ii) Find P(use App B | uses App A) — i.e., given that a user uses App A, what's the probability they also use App B?
(iii) Are using App A and App B independent events? Justify with a calculation.

[3 Marks]

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a). A university has three departments D_1, D_2, and D_3, which contain 40%, 35%, and 25% of the total students, respectively. The probabilities that a student from these departments receives a scholarship are 0.10, 0.20, and 0.30, respectively. If a randomly selected student is found to have received a scholarship, determine the probability that the student belongs to department D_2. Show the necessary steps.

[3 Marks]

b). A discrete random variable X has the following probability distribution:

X	0	1	2	3	4
P(X = x)	0.10	0.25	K	0.20	0.15

(i) Find the value of k that makes this a valid probability distribution.
(ii) Find $P(X \leq 2)$ and $P(X > 2)$.
(iii) Find the expected value E(X) and variance Var(X).

[3 Marks]

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a. In a biological nursery research study, the growth of young plants after a fixed period is normally distributed with a mean of 100 cm and a standard deviation of 15 cm. **[3 Marks]**

(i) What is the probability that a randomly selected plant has a growth above 130 cm?
(ii) If a plant is selected at random, what is the probability that its growth falls between 85 cm and 115 cm?
iii) If a plant is selected at random, what is the probability that its growth falls below 120 cm?

b. A call center finds that 8% of incoming calls result in a customer complaint being escalated to a supervisor. A shift handles 25 calls.
(i) Find the probability that exactly 3 calls are escalated.
(ii) Find the probability that at most 2 calls are escalated.
(iii) Find the probability that at least 1 call is escalated.

[3 Marks]

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a). A company wants to predict whether a customer will Purchase a product based on two attributes: Age Group and Income Level. The following training data are available:

Age Group	Income	Purchase
Young	High	Yes
Middle	Low	Yes
Senior	High	No
Young	Low	No
Middle	Medium	Yes
Senior	High	No
Young	Medium	Yes
Middle	Medium	Yes
Senior	Medium	No
Young	High	No

Using the Naive Bayes Theorem, predict whether a new customer with Age Group = Young and Income = Medium will Purchase or Not Purchase the product. Show the calculation of the prior and conditional probabilities and clearly state the final prediction. **[4 Marks]**

b). A class has 50 students where each student has a probability p of passing an exam. The exam is conducted with a proctor so that students cannot copy from each other. What is the expected number of students who pass the exam? Can we use the Binomial distribution to model this situation? If so, find the probability that exactly 10 students pass the exam. Otherwise explain why the Binomial distribution cannot be used in this case. **[2 Marks]**

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a). The lifetime (in years) of a certain sensor component follows $f(x)= 2kx$ for $0 \le x \le 5$, and 0 otherwise. **[3 Marks]**

(i) Find the value of k that makes this a valid PDF.

(ii) Find $P(2 < X < 4)$.

(iii) Find E(X) and Var(X).

b).The time customers spend on a telecom company's support call has a (non-normal, right-skewed) mean of 8 minutes with a standard deviation of 3 minutes. A random sample of 100 calls is taken. **[3 Marks]**

(i) Can you still apply the Central Limit Theorem here? Justify why or why not, given the population isn't normal.

(ii) Find the probability that the sample mean call time exceeds 8.5 minutes.

(iii) How would your answer to (i) change if the sample size were only n = 15 instead of 100?